Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/25/2017
PI Name:	Rickus, Jenna Ph.D.		
Project Title:	Sporesat 2 Mission: Investigating Biophysical Mechanisms of Plant Gravisensing Using a Lab-On-A-Chip Approach (CoPI=Rickus)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 SALMON--Astrobiology & FSB NNH08ZDA009O-SCMAFSB
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Key Personnel Changes/Previous PI:	n/a		
COI Name (Institution):	Roux, Stanley Ph.D. (Co-PI: University of Texas, Austin)		
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Task Description:	Ed. NOTE (June 2015): Continuation of "Sporesat Mission: Investigating biophysical mechanisms of plant gravisensing using a lab-on-a-chip approach"--grant NNX11AF48A for SporeSat 2. SporeSat is a Small Complete Mission based on a Mission of Opportunity-2 nanosatellite designed for real time measurement of calcium signaling and maintenance of variable gravity treatments during flight. The impetus for SporeSat is to gain a deeper knowledge of the molecular and biophysical mechanisms for plant gravity sensing by studying the activation of cell gravity-sensing and electrophysiological signaling in the single-cell model system, Ceratopteris richardii, using a novel 'lab-on-a-chip' microsensor technology platform called a CEL-C advanced bioCD. The science and technical objectives as outlined in the original SporeSat proposal: Science Objectives: 1) Determine gravity thresholds for Ca2+ channel activation in wild-type spores; 2) Observe Ca2+ current response of gene knockout mutants. Technology Objectives: 1) Design and fabricate the CEL-C advanced bioCD; 2) Integrate bioCD with data acquisition hardware and ground-based validation experiments; 3) Develop and integrate centrifuge hardware; 4) Operate bioCD in space-relevant environment; 5) Conduct spaceflight experiments. In 2013, SporeSat was adapted to a 2 flight mission strategy with flight 1 focused on the technology objectives and flight 2 focused on the scientific objectives. In April 2014, SporeSat-1 was deployed from a SpaceX Falcon 9 launched from Cape Canaveral as part of CRS-3. SporeSat-1 was successfully commanded and science data downloaded. However, the OLED light source failed and one of the three bioCDs overheated (40C) above its target set point (29C). These issues were addressed by the Ames engineering team via a Failure Review Board process. SporeSat1 also faced other technical challenges related to the biology. At the time of SporeSat1, the strain of spore that had been used for all prior gravity modulated calcium measurements, Hn-n, was not physically available anywhere in the world. Instead the next best strain, Rn5, was used but was limited by a lack of prior calcium data as well as reduced germination rates due to age of spore batch (~5 years). In addition, the available methods of generating the desired caclium mutants were not reliable. Since that time, Co-Principal Investigator Stan Roux and Mari Salmi produced new batches of Hn-n spores and new methods of genetic manipulation of spores became available. Among the improvements made prior to SporeSat2 was an improved bioCD enclosed assembly (EA) wall design. The first SporeSat EA wall design contained an agar volume that was determined to be insufficient for the fern spores during storage. The EA walls and spore wells were redesigned to increase the agar volume available for each spore and was used in SporeSat-1 flight and ground controls. The new design, while an improvement for the spores, created an unintentional electrical coupling of the Ca2+ sensors. In addition this change in wall design may have contributed to the moisture leakage that was involved in OLED and temperature control failure. We therefore created a design upgrade for SporeSat-2 that retained a higher agar volume but that electrically isolated the spore wells and Ca2+ electrode pairs. In November of 2014 SporeSat2 began and was parsed into a Phase I (ground) and Phase II (flight). The primary goal of Phase I was to demonstrate that the Hn-n and/or Rn5 fern spores exhibit a reliable and repeatable Ca2+ current that is measureable by the bioCD during germination under ideal ground conditions. In this report we focus on the major accomplishments achieved during the Jan 2015 – May 2016 period which encompassed SporeSat2 Phase 1. The SporeSat-2 experimental paradigm and geometry is different from any previous experimental set ups. As in any biosensor experiment, the Ca2+ ISEs are sampling at a particular location in space and time, a dynamic calcium current that exists across the 3D surface of the fern spore. To assist in development of the biological model based on sensor recordings, we have developed a 3D simulation model that predicts the 3D calcium concentration profiles in the SporeSat wells as well as the sensor recordings over time based on the biological flux from the fern spore. We will therefore be able to compare our experimental data with our predictions for different conditions (angle off gravity, under rotation) as well as for different calcium mutants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Astronaut health in long term space missions is likely to be dependent on and benefit from plants. On Earth we depend on plants for food, clean air, and waste treatment. The same will be true in space. If we as humans are to achieve the kind of space exploration and habitation that we all imagine and dream of, we are going to need plants up there with us. Yet on the International Space Station (ISS), gravity is less than one one-thousandth of the gravity here on Earth. On the Moon, it is about 17% of Earth's gravity. In addition, there are astonishing similarities in signaling within cells across species. As in plants, calcium plays important roles in human cell development, function, and dysfunction. Understanding gravity sensing in plant cells is likely to give clues to gravity sensing in other organisms including humans. In addition, the technology that we developed for these experiments establishes a platform to ask other questions regarding human cell responses on Earth and in space. We are already looking at how we can adapt the technology to study pancreatic beta cells and diabetes.
	Elucidation of the mechanisms by which living cells sense and respond to variable gravity is fundamentally important to the advancement of human space exploration, and calcium is an important mediator of cell signal transduction in all organisms. In plants, environmental factors as diverse as temperature, light, touch, and gravity induce an increase in cytoplasmic calcium concentration, and this, in turn, initiates signalling cascades leading to adaptive physiological changes. Calcium signalling has long been known to play a role in both the earliest stage of gravity perception and the transmission of the signal in gravity-directed polar development. Spores of the fern Ceratopteris richardii are a valuable system for studying cellular gravity response. Each spore is a single cell, and before the first division of the cell, the orientation of cellular polarity is set in response to the force exerted on the cell by gravity. The mechanism of gravity response in the spore cell includes the transport of calcium into and out of the cell. Soon after the spore germination process begins, calcium is transported out of the cell, against its concentration gradient, on the sides and top of the spore, with peak efflux of calcium at the top of the spore. At the same time, calcium is transported into the spore at the bottom. This polarized orientation of calcium transport is directly linked to the force and orientation of gravity as demonstrated in lab-on-a-chip experiments that monitored the calcium differential while the spores were reoriented relative to gravity, and also while the spore experienced varying g-forces onboard parabolic flight. The direction of calcium transport in the spore cell results in a cell polarization that ultimately

Task Progress:	correlates with the direction of rhizoid growth. Measurement of the gavity-induced calcium polarization over longer developmental time periods is necessary to assess the effect of gravitational magnitude and to measure the minimum threshold of gravity necessary to set the cell's developmental polarity. In order to measure polarized calcium signals from fern spores over long periods of germination time (hours) in varying gravity, we designed a second-generation lab-on-a-chip device that supports on-disk electrical measurement during rotation, referred to here as a bioCD. The bioCD can be operated autonomously under continuous rotation or while stationary during space flight in a free-flying nanosatellite, known as SporeSat. SporeSat (5.5 kg, 10x10x34 cm) can be carried to space as a secondary payload and inserted into low Earth orbit at altitudes ranging from 350 to 750 km). It builds on the design of previous small satellite missions to measure the effects of microgravity on living organisms. In this work period we demonstrated the ability to calibrate the calcium response of single cell fern spores subject to hypergravity while on Earth in a single experiment. The bioCD with supporting hardware measures calcium ion concentration ratios indicative of the polarization of transcellular ion currents in single cells under varying artificial gravity created by rotation. We optimized the function of the on chip amplifiers to reduce noise, implemented statistical methods for rigorously identifying the positive biological signals, and improved the well quality control procedures. These improvements enabled the reliable measurement of gravity induced calcium signals on a single bioCD. We demonstrated the system's function as an autonomous programmable experiment on Earth by measuring calcium concentration differentials around single-cell fern spores. We measured the first calibration of these differentials as a function of gravity magnitude during a full 24 hours of rotation following light-induced germination. Long-term hypergravity stimulation resulted in an electrochemical signal that is similar in charge orientation differential to that previously reported in response to very short term hypergravity stimulation. This gravitational force-induced cellular signal is significantly different from the negative controls. The magnitude of this signal increases with increasing net gravity. The novel and most valuable aspect of the SporeSat free-flying nanosatellite design is that it will provide multiple gravitational levels for spores during development in microgravity. The most common methodologies for experiments aimed at understanding cell, organ, or whole organism mechanisms of gravity perception involve randomization of the gravity vector on Earth. While these experiments are useful and necessary, the SporeSat nanosatellite provides a platform for a true microgravity experiment with simulated gravity controls. In addition, the biological system used provides important electrochemical information about the mechanism a single cell uses to detect the subtle force of gravity. The SporeSat bioCD can now be used as an important tool to evaluate the contribution of individual gene targets to this gravity-induced electrochemical signal. It can also be used to determine the minimum gravitational force needed to activate a single cell response.
Bibliography Type:	Description: (Last Updated: 04/26/2017)
Articles in Peer-reviewed Journals	Park J, Salmi ML, Wan Salim WW, Rademacher A, Wickizer B, Schooley A, Benton J, Cantero A, Argote PF, Ren M, Zhang M, Porterfield DM, Ricco AJ, Roux SJ, Rickus JL. "An autonomous lab on a chip for space flight calibration of gravity-induced transcellular calcium polarization in single-cell fern spores." Lab Chip. 2017 Mar 14;17(6):1095-103. http://dx.doi.org/10.1039/c6lc01370h ; PubMed PMID: 28205656 , Mar-2017